Abstract

This report presents the revised preliminary investigation for an MSc dissertation project proposing a three-stage, data-driven pipeline for automated image captioning in the grocery retail domain. The central research problem is the prevalence of hallucination in Large Vision-Language Models (VLMs) — the generation of plausible but visually unverified content — which renders current end-to-end generative models unsuitable for operationally critical deployment. Prior tag-guided captioning approaches, such as Tag2Text and Semantic Compositional Networks, have demonstrated that semantic tags can meaningfully guide caption generation [1], [2], yet none have addressed domain-specific retail challenges or incorporated unsupervised scenario routing as a structural constraint mechanism. This project proposes to decompose the captioning task into three layers: (1) a Deterministic Tagging Layer employing a hierarchical four-tier ontology, (2) an Unsupervised Routing Layer that autonomously discovers operational scenarios from tag co-occurrence patterns, and (3) a Conditional Captioning Layer using structured prompt injection with Explicit Absence Encoding. Ground truth tags and reference captions are produced via an LLM-Assisted Semi-Automated Annotation workflow with student-led review, a practical and academically defensible approach for MSc-level research. The available dataset, sourced from HuggingFace, provides rich file-path-embedded metadata — campaign type, retail brand, and product category — that directly informs the Automated Dynamic Dictionary without requiring external data sources. The system will be evaluated using CHAIR, POPE, CIDEr, SPICE, and a custom Retail Actionability Score (RAS) on a small-scale student evaluation panel.

1. Introduction

The automated interpretation of visual data in retail environments has become an increasingly important capability, with applications spanning shelf audit automation, planogram compliance, accessibility services, and smart product cataloguing. Recent advances in deep learning and vision-language pretraining have produced powerful image captioning systems [3], yet a fundamental tension persists between generative fluency and factual fidelity.

Large Vision-Language Models trained to maximise token-level likelihood are inherently probabilistic. In retail environments — where the precise identification of product states, fixture types, and operational anomalies is required — this probabilistic tendency leads to hallucination: the generation of descriptions referencing objects, promotions, or stock conditions that do not appear in the image but are statistically plausible given training priors [4]. A system that describes a fully-stocked shelf when a product line is absent, or fabricates promotional signage based on learned associations, poses unacceptable risks in operational deployment.

This project addresses this problem through a structural constraint approach: by extracting a verified set of visual tags prior to generation and conditioning the VLM on both confirmed visual evidence and confirmed absences, the model's generative entropy is reduced and hallucination is structurally discouraged. The project further introduces an unsupervised routing mechanism that translates the tag set into an operational task context autonomously, without reliance on manual expert rule matrices. Both contributions are absent from the existing tag-guided captioning literature [1], [2], [5].

2. Research Objectives and Research Questions

The project is organised around one primary and three subsidiary research questions, each addressed by a specific experimental component.

Primary Research Question (PRQ): Can a data-driven, tag-constrained pipeline — comprising hierarchical deterministic tagging, unsupervised scenario routing, and explicit absence encoding — significantly reduce object hallucination and increase operational actionability in VLM-generated retail image captions compared to an unconstrained end-to-end baseline?

RQ1: Does unsupervised clustering on tag co-occurrence patterns produce semantically valid and operationally coherent retail scenario groupings without manual expert input?

RQ2: Does Explicit Absence Encoding statistically reduce object hallucination rates (CHAIR) and improve object existence accuracy (POPE) relative to tag-injection-only and unconstrained baselines?

RQ3: Does persona-conditioned captioning via the automated routing layer produce captions rated as more operationally actionable (RAS) than flat-tagged or unconstrained alternatives?

3. Literature Review

The proposed system sits at the intersection of image captioning, semantic tag-guided generation, VLM hallucination mitigation, and domain-specific visual understanding. This section reviews the most relevant prior work and identifies the specific gaps this project addresses.

3.1 Image Captioning: From Statistical to Neural Approaches

Early image captioning systems discovered statistical correlations between image features and descriptive keywords to produce keyword-based annotations of novel images [6]. The field has since been transformed by deep learning, with transformer-based encoder-decoder architectures and large-scale pretraining becoming the dominant paradigm [3]. Comprehensive surveys of deep learning captioning approaches consistently identify hallucination, missing context, and domain generalisation as the primary unresolved challenges in the field [3]. Surveys of earlier generation-based and retrieval-based approaches further established the evaluation metrics and benchmark datasets — including COCO and Flickr30K — that remain in standard use [7].

3.2 Tag-Guided Image Captioning

The use of image-level semantic tags as generative guidance has a well-established lineage. You et al. proposed a semantic attention mechanism that selectively attends to detected concept proposals during recurrent generation, demonstrating significant improvements on standard benchmarks [8]. The Semantic Compositional Network (SCN) extended this by using tag detection probabilities to compose the weight matrices of an LSTM network, making generative parameters explicitly tag-dependent and showing that soft tag probability weighting consistently improves caption quality [2]. HAAV treats image tags as one type of augmented view alongside visual grid features, aggregating heterogeneous encodings hierarchically and reporting substantial CIDEr improvements on COCO and Flickr30K [9]. ViTCAP introduced a Concept Token Network that predicts semantic concept tokens via an auxiliary classification task and injects them into an end-to-end transformer captioning model, achieving competitive performance without a separate region detector [5]. TagAlign (Liu et al., 2023) demonstrates that using an LLM to automatically parse objects and attributes from image captions — and training the vision encoder with a multi-tag classification loss against these parsed tags — improves fine-grained image-text alignment by an average 5.2% on semantic segmentation benchmarks, showing that LLM-aided vocabulary definition produces higher-quality tag supervision than rule-based keyword-extraction methods such as NLTK [10]. TIPCap (Wang et al., 2024) proposes an interactive prompt module for weakly supervised captioning in which user-specified object tags serve as corrective signals injected into the decoder at inference, demonstrating that targeted tag injection can directly correct factual errors in generated captions without model re-training [11].

Tag2Text is the most directly related prior work, proposing an image-tag-text generation scheme in which a dedicated tag recognition decoder guides a vision-language model's captioning output across general-domain imagery [1]. This project distinguishes itself from Tag2Text on three key dimensions: (i) a hierarchically structured retail ontology constrains tags to operationally meaningful, industry-aligned categories; (ii) an unsupervised routing layer translates the tag set into an operational task context before generation, a step entirely absent from Tag2Text; and (iii) Explicit Absence Encoding converts confirmed non-detections into positive generative constraints, a mechanism not present in any prior tag-guided captioning work. Singh (2025) validates the hierarchical attribute-first generation paradigm in an e-commerce context, reporting that embedding model-predicted product attributes into a structured generation prompt reduces hallucination from 12.7% to 7.1% (a 44.5% relative improvement) while reducing generation latency 3.5×, further supporting the principle that tag-constrained prompt structures are more effective than unconstrained end-to-end generation in domain-specific settings [12].

3.3 Domain-Specific and E-Commerce Visual Description Generation

While general-domain captioning research has demonstrated the value of tag-guided generation, a smaller but increasingly relevant body of work addresses visual description generation in commercial and retail domains. OPAL (Optimized Preference-Based AI for Listings; Zhang et al., 2025) proposes a framework for generating schema-compliant item descriptions from product images using a fine-tuned MLLM, addressing the modality gap between visual and textual representations through two data refinement methods: MLLM-Assisted Conformity Enhancement for structured schema alignment and LLM-Assisted Contextual Understanding for fine-grained visual detail capture, demonstrating consistent improvements in description quality and schema completion rates over baselines on real-world e-commerce datasets [13]. The TUNA framework (Qi et al., 2024) demonstrates that injecting retrieval-augmented object tags — covering object names, attributes, and novel entities from a large-scale external datastore — into MLLM prompts consistently reduces mention of non-existent objects, improves identification of out-of-distribution entities, and increases attention to visual details, outperforming baselines on 12 VQA and multimodal benchmarks; crucially, their experiments confirm that tags provide cleaner and more object-specific information than retrieved captions, and that tag-grounded generation is therefore more effective than sentence-level retrieval for domain-specific tasks [14]. The VRAP framework (Rivera et al., 2024) independently validates that structured tags — including objects, attributes, and relationships extracted via pretrained visual encoders and scene graph parsers — when incorporated into LLM prompts via retrieval-augmented prompting achieve state-of-the-art performance on VQAv2, GQA, and VizWiz benchmarks, with offline tag pre-generation reducing inference latency by 40% [15]. Both OPAL and TUNA confirm that domain-specific schema constraints applied through structured prompt injection provide measurable hallucination suppression and description quality improvements. However, none of these systems address the multi-context operational diversity characteristic of grocery retail environments — where a single dataset spans fundamentally different task types (inventory audits, safety inspections, promotional compliance) — nor do they incorporate unsupervised task-context routing or explicit absence-signal mechanisms as structural constraints.

3.4 Hallucination Mitigation in VLMs

Hallucination in VLMs arises from the tension between pixel-level image evidence and the model's learned inductive bias over plausible image-text associations [4]. ViECap addresses this in zero-shot captioning by constructing entity-aware hard prompts that direct LLM attention toward visually confirmed entities, demonstrating reduced object fabrication and improved cross-domain transfer [4]. VIVO addresses the related problem of novel object captioning by pretraining on large-scale image-tag pairs using a Hungarian matching loss, enabling visual-tag alignment without paired caption data [16]. While these approaches confirm that structural input constraints can reduce hallucination, none address the specific failure mode of fabricated operational states (e.g., invented promotional offers, false stock-level descriptions) in retail VLM deployment.

3.5 Research Gap

The reviewed literature confirms three converging findings: tag-guided captioning improves factual grounding in general domains [1], [2], [5]; retrieved or predicted object tags, when injected as structured prompts, measurably reduce hallucination in MLLMs [14], [15]; and schema-constrained generation in e-commerce settings produces more accurate and operationally aligned outputs than unconstrained end-to-end generation [12], [13]. However, no existing work: (i) applies a hierarchically structured retail ontology aligned to GS1 GPC industry standards as the tagging backbone; (ii) introduces an unsupervised tag co-occurrence routing layer to autonomously assign distinct operational task contexts to images at inference time; or (iii) implements Explicit Absence Encoding — converting confirmed non-detections into positive generative constraints — as a structural hallucination suppression mechanism. The closest analogues in domain-specific captioning (OPAL, Singh 2025) address schema compliance in single-product e-commerce listing generation but operate on homogeneous product imagery without any routing requirement and without absence-signal mechanisms. The tag-grounded frameworks TUNA and VRAP improve general-domain MLLM performance via tag injection but neither employ a task-routing layer nor target the specific operational failure modes of grocery retail deployment. This project addresses all three gaps within a unified, compute-efficient pipeline designed specifically for multi-context grocery retail environments.

4. Proposed Methodology

The system implements a three-stage pipeline — Dynamic Context Tagging → Unsupervised Routing → Conditional Captioning — in which each stage produces a structured output consumed by the next, enabling independent evaluation and targeted ablation.

4.1 Stage 1: Dynamic Context Tagging (Deterministic Layer)

Stage 1 produces a verified, structured tag set T for each input image I, including only visual elements that have crossed defined confidence thresholds. Four extraction modules correspond to the four levels of the hierarchical ontology.

L1 (Scene/Zone): CLIP (ViT-L/14) performs zero-shot classification of the overall spatial context using text prompts such as "a photo of a retail trading floor aisle", "a photo of a supermarket checkout zone", and "a photo of a retail stockroom". This approach leverages CLIP's broad visual-language alignment without requiring a separately trained scene classifier or labelled training images. **L2 (Fixture Category)**: Grounding DINO — an open-vocabulary detection model capable of localising objects from free-text descriptions — identifies planogram fixture types (e.g., Gondola Shelving, Endcap Display, Pallet Stack, Refrigerated Bay) without any bounding-box annotation or domain-specific fine-tuning. This replaces the originally planned YOLOv8 fine-tuning approach, which was determined to be infeasible given the absence of bounding-box annotated training data and the time constraints of an MSc project. **L3 (Product Type)**: CLIP (ViT-L/14) performs zero-shot matching against a GS1 Global Product Classification (GPC) Brick-level vocabulary — the internationally standardised product classification schema used across UK grocery retail supply chains — augmented by the Automated Dynamic

Dictionary (see Section 4.1.1). If CLIP zero-shot precision on fine-grained product categories falls below an acceptable threshold during Phase 1 validation, a lightweight linear probe classifier will be trained on frozen CLIP embeddings of the 300–400 annotated images, providing a data-efficient fallback that still requires no bounding-box annotations. **L4 (Attribute/State)**: PaddleOCR extracts visible text (price tags, promotional signage), and a CLIP-based zero-shot attribute classifier identifies operational states (Empty Shelf, Damaged Product, On Promotion) via prompt matching against state-description text templates.

Tags are classified into two tiers: **Hard Tags** (confidence > 0.85) serve as absolute generative constraints, and **Soft Tags** (confidence 0.60–0.85) are passed as probabilistic hints. Tags scanned for but not detected above either threshold are recorded as **Confirmed Absences**, forming the input to Stage 3's Explicit Absence Encoding mechanism. The initial confidence thresholds (0.85 for Hard Tags; 0.60 for Soft Tags) are treated as starting values derived from standard CLIP zero-shot classification practice; they will be empirically re-calibrated on the validation subset (approximately 150 images) using F1 score maximisation against the human-reviewed ground truth, and the final calibrated values will be reported in the dissertation.

4.1.1 Automated Dynamic Dictionary via File-Path Metadata

A key innovation of Stage 1 is the Automated Dynamic Dictionary, which adapts the L3 vocabulary to temporal and contextual factors without manual database maintenance. Rather than relying only on image EXIF data, this project leverages the rich metadata embedded in the dataset's file-path structure, which encodes the following fields:

- **Campaign / Season**: e.g., `Halloween2024`, `FullStores` — directly indicates the thematic context and seasonal period
- **Retail Brand**: e.g., `Tesco`, `Waitrose`, `Asda` — provides brand-specific store format and product range priors
- **Product Category**: e.g., `Food - Floral`, `Branding Signage` — provides a coarse L2/L3 classification prior extracted without any model inference
- **Store / Location**: e.g., `Asda Killingbeck` — enables store-format-level context (e.g., discount vs. premium positioning)
- **Date metadata**: year/month/day from image capture records

A lightweight orchestrator LLM (e.g., GPT-4o-mini via API) is prompted with the parsed path metadata to generate a probabilistic list of expected contextual keywords. For example, the path `train/Halloween2024/Tesco/Food - Floral/` produces a dictionary enriched with terms such as Pumpkin, Cobweb, and Witch Hat for L3 matching. CLIP performs zero-shot matching against this dynamically generated vocabulary; matches above a calibrated threshold (≥ 0.65) are added to T as Probabilistic Tags. This approach turns an existing dataset structural feature — the file-path hierarchy — into a zero-cost contextual signal, requiring no additional metadata collection.

4.1.2 Hierarchical Retail Ontology: L1–L4 Tag Schema

Ontology Design Methodology. A core methodological challenge in constructing the L1–L4 ontology is that the controlled vocabulary — particularly L1 scene types and L2 fixture types — cannot be reliably derived from file-path metadata alone, which encodes campaign, brand, and coarse product category but does not describe the spatial layout or fixture structure of a scene. Imposing vocabulary by assumption without domain expertise would risk producing categories that do not map onto the actual visual content of the dataset. To address this, the L1–L2 vocabulary is derived via a systematic two-pass data-driven induction process rather than specified arbitrarily.

In **Pass 1 (Corpus Induction)**, GPT-4o Vision is applied to a stratified random sample of 100 images drawn from the dataset. For each image, GPT-4o is prompted with the following instruction: *"Describe in noun-phrase form only: (i) the type of retail space or zone visible in this image; (ii) any display structures or fixtures present; (iii) the broad product categories visible; and (iv) any observable operational states such as stock levels, promotional signage, or physical anomalies."* The 100 free-form noun-phrase responses are then aggregated, and GPT-4o is prompted to cluster recurring terms into a candidate hierarchical taxonomy. This produces an initial vocabulary directly grounded in the actual visual content of the dataset, rather than in generic retail assumptions. In **Pass 2 (Standard Alignment)**, the candidate vocabulary is cross-referenced with established industry frameworks: L2 fixture types are aligned with retail visual merchandising literature; L3 product types are mapped directly onto GS1 Global Product Classification (GPC) Brick nomenclature; and L4 operational state attributes are derived from the operational research questions guiding this project — specifically, the dimensions of stock level, promotional state, planogram compliance, safety condition, and shelf cleanliness. The finalised vocabulary is simultaneously empirically grounded in the specific dataset, terminologically consistent with industry standards, and operationally relevant to the project's evaluation design. The full induction procedure and the final vocabulary list for each level will be documented in the dissertation methodology chapter and included as an appendix to ensure transparency and reproducibility.

The four-tier ontology defines the controlled vocabulary for all tag extraction modules. The table below presents the finalised vocabulary at each level, as derived through the two-pass induction process described above. Critically, the L3 level adopts GS1 GPC Brick nomenclature directly as its product-type vocabulary, and the L4 level adopts the five operational state dimensions derived from the project's research questions. Because GS1 GPC is the standardised classification schema used across UK grocery retail supply chains, L3 tags generated by this system carry industry-standard product terminology that is compatible with downstream retail management systems (e.g., shelf management software, inventory databases) without requiring terminology translation. This alignment is both a methodological design choice and a practical contribution. The associated detection method for each level is also noted.

Level	Category	Finalised Vocabulary (Two-Pass Induction)	Detection Method

L1	Scene / Zone	Ambient Aisle, Fresh Produce Section, Chilled / Refrigerated Aisle, Frozen Section, Checkout / POS Area, Promotional Display Area, Storage / Backroom	CLIP zero-shot classification
L2	Fixture Type	Gondola Shelf, Wall Shelf, End Cap, Open Refrigerated Case, Freezer Door Unit, Dump Bin, Freestanding Display Unit (FSDU), Checkout Counter, Pallet / Floor Stack	Grounding DINO open-vocabulary detection
L3	Product Type (GS1 GPC Brick)	Ambient Grocery – Canned Vegetables, Fresh Produce – Root Vegetables, Confectionery – Chocolate, Beverages – Carbonated Soft Drinks, Health & Beauty – Shampoo (full Brick list sourced from GS1 GPC Browser)	CLIP zero-shot + Dynamic Dictionary (GPT-4o-mini)
L4	State / Attribute	Stock Level (Full / Partial / Empty), Promotional State (Active Promo / None), Planogram Compliance (Compliant / Non-compliant), Safety Condition (Clear / Hazard Present), Shelf Cleanliness (Clean / Untidy / Damaged Packaging)	PaddleOCR + CLIP zero-shot attribute classifier

When L3 tags are injected into Stage 3 caption prompts as Hard or Soft Tags, GS1 GPC Brick names are used verbatim (e.g., `Beverages – Carbonated Soft Drinks`). This means the generated captions directly reference standardised retail product vocabulary, which is a practical advantage over free-form product naming. If CLIP zero-shot precision on fine-grained L3 categories falls below an acceptable threshold during Phase 1 validation, a lightweight linear probe classifier trained on frozen CLIP (ViT-L/14) embeddings of the annotated subset will be used as a fallback — requiring no additional bounding-box labels and feasible within MSc project resource constraints.

4.2 Stage 2: Unsupervised Data-Driven Routing (Logic Layer)

Stage 2 transforms the tag set T into an operational Task Context K, comprising a Persona (the intended recipient of the caption) and a Primary Goal (the information the caption must convey), derived autonomously from data rather than from manually engineered rules.

Tags at levels L1 (Scene/Zone) and L4 (State/Attribute) are selected as the clustering feature space, as these two levels most directly capture the *operational scenario* of an image — specifically, where in the store something is happening (L1) and what condition or anomaly is present (L4). L2 (fixture type) and L3 (product type) provide fine-grained object identity but are less discriminative for routing to a macro-level operational persona. Each image is encoded as a multi-hot sparse vector over the combined L1 + L4 tag vocabulary, resulting in a high-dimensional but sparse feature matrix. Unsupervised clustering (K-Means for a specified scenario count; DBSCAN for density-based separation) is applied to identify statistically significant co-occurrence patterns. The optimal K for K-Means is selected by maximising the

Silhouette Score and minimising the Davies-Bouldin Index across K = 3 to 8. Once clusters are identified, the orchestrator LLM (GPT-4o-mini) is prompted with each cluster's tag distribution and asked to assign a Persona and Primary Goal, producing an Automated Routing Matrix. Representative example outputs include:
`{Trading Floor + Out of Stock + Price Tag Missing} → Inventory Manager → "Identify the out-of-stock product by its shelf adjacency context and note the missing price label"`
and
`{Trading Floor + Product Damaged + Planogram Violation} → Safety Officer → "Describe the hazard type, location, and severity for incident logging"`
. If clustering is unstable or produces non-interpretable groups, three manually seeded scenario types (Inventory, Safety/Compliance, Merchandising) will be used as a structured fallback, with the clustering failure documented as a negative empirical finding.

4.3 Stage 3: Automated Conditional Captioning (Generative Layer)

Stage 3 generates the final natural language caption conditioned on the verified tag set T and the task context K using a Mixture of Prompts (MOP) strategy. The prompt is assembled as a structured object with zero user intervention, composed of: (i) Persona and Task Instruction from Stage 2; (ii) Hard Constraint Tags as verified visual facts; (iii) Soft Hint Tags with explicit uncertainty notation; and (iv) Explicit Absence Assertions (e.g., `Price_Tags: scanned, count = 0`).

The Explicit Absence Encoding mechanism converts negative detection outcomes into positive assertions, preventing the VLM from filling unaddressed features with prior-driven fabrications — a structural extension of the entity-aware prompting approach demonstrated in ViECap [4]. For high-stakes tasks, a Self-Correction loop is activated: the VLM is asked to audit its own caption against the Hard Constraint Tag list and flag any mentioned object absent from the verified set.

Given personal GPU constraints, the generative VLM is **LLaVA-1.5-7B or Qwen-VL-Chat (7B)**, selected for strong instruction-following capability within manageable VRAM requirements. For final evaluation runs requiring higher throughput, KCL HPC resources will be requested. LLM API access (GPT-4o-mini) is used for the orchestrator functions in Stages 1 and 2, keeping local inference load focused on the vision models.

5. Data Plan

The dataset has been pre-secured from HuggingFace, comprising approximately 10,000 grocery retail images across multiple UK retail brands (Tesco, Waitrose, Asda, and others) with campaign-level, brand-level, and category-level organisation encoded in the file-path hierarchy.

5.1 Preprocessing

A Laplacian variance filter will reject images below a sharpness threshold (calibrated empirically on a random 100-image pilot sample), targeting retention of at least 9,000 usable images. File-path metadata will be parsed and stored as a structured JSON sidecar for each image. Privacy-related preprocessing — face detection via RetinaFace and Gaussian blur — will be applied before any VLM inference. The 300–400 annotated images will be divided into two non-overlapping subsets: a **validation subset** (approximately 150 images), used for confidence threshold calibration, iterative prompt tuning, and system development; and a **held-out test subset** (approximately 150–200 images), reserved exclusively for final ablation evaluation and metric reporting. No model parameters, confidence thresholds, or prompt structures will be adjusted using held-out test images, ensuring a clean separation between development and evaluation to prevent inadvertent leakage.

5.2 LLM-Assisted Semi-Automated Annotation with Student Review

As the researcher is not a domain expert, ground truth annotation follows a practical LLM-assisted workflow rather than a pure manual expert process. A stratified sample of **300–400 images** (selected to represent the major L1 zones, campaign types, and operational states present in the dataset) will be annotated as follows:

- **Tag Ground Truth**: The researcher directly reviews each sampled image against the L1–L4 ontology schema through visual inspection to produce validated tag labels. GPT-4o (with image input) is used as a suggestion generator to accelerate this process — its proposed tags are treated as candidates requiring human confirmation or correction, not as authoritative annotations. This human-verification-first design ensures that the tag ground truth used for Retail-CHAIR evaluation is independently established from the LLM family used in caption generation, thereby mitigating the circular reference risk inherent in fully LLM-automated pipelines. A 50-image subset will be independently reviewed by a second student annotator to compute inter-rater reliability (Cohen's kappa), providing a quantitative quality estimate of annotation consistency that will be reported in the dissertation.
- **Reference Captions**: For a subset of approximately 150–200 images (used for CIDEr and SPICE computation), GPT-4o is prompted to generate 2–3 candidate reference captions per image. The researcher reviews and selects or lightly edits the most factually accurate candidate. It is explicitly acknowledged that reference captions sourced from the same model family (GPT-4o) as the LLM-as-Judge evaluation introduces a potential consistency bias in the CIDEr/SPICE scores; these metrics are therefore treated as supplementary caption quality indicators only. Retail-CHAIR and POPE — both computed against the independently human-reviewed tag ground truth — serve as the primary hallucination metrics and are not subject to this dependency. This limitation will be discussed explicitly in the dissertation's evaluation chapter.

5.3 Dataset Statistics Reporting

The dissertation will report: (i) distribution of images across parsed campaign types and brands; (ii) frequency of L4 attribute states (Empty, Damaged, On Sale) in the annotated subset; and (iii) number and proportion of images per cluster discovered in Stage 2, to contextualise routing validity.

6. Evaluation Framework

The evaluation framework is structured to answer each research question through a combination of quantitative pipeline metrics, standard captioning benchmarks, and small-scale human evaluation.

6.1 Quantitative Metrics

Stage 1 — Tag Detection Quality: Precision and Recall computed against the 300–400-image annotated ground truth. Dynamic Tag Recall: precision and recall of the LLM-generated Dynamic Dictionary entries against the annotated seasonal image subset.

Stage 2 — Routing Validity (RQ1): Silhouette Score and Davies-Bouldin Index for cluster cohesion and separation across K = 3 to 8 scenarios. Routing Appropriateness: a 20–30-image sample per cluster is reviewed by the researcher and one peer, assessing whether the LLM-assigned persona logically reflects the cluster's visual content (reported as agreement rate).

Stage 3 — Hallucination Reduction (RQ2): The primary hallucination metric is **Retail-CHAIR**, a domain-adapted variant of the standard CHAIR (Caption Hallucination Assessment with Image Relevance) metric. Standard CHAIR checks whether object nouns in a generated caption appear in the COCO reference object vocabulary; Retail-CHAIR replaces this vocabulary with the project's L1–L4 ontology tags. The concrete implementation is as follows: (1) for each held-out test image, the manually annotated L1–L4 tag set constitutes the ground-truth reference vocabulary; (2) noun phrases are extracted from the generated caption using spaCy dependency parsing; (3) each extracted noun phrase is matched against the ground-truth tag set using exact string matching and GS1 GPC synonym equivalences; (4) CHAIR_i (image-level: fraction of images containing at least one hallucinated object mention) and CHAIR_s (sentence-level: fraction of sentences containing a hallucinated mention) are computed. This implementation requires no retail domain expertise beyond what is already encoded in the annotation guidelines: the L1–L4 ontology itself constitutes the domain knowledge proxy, and its construction draws on publicly available GS1 GPC standards rather than subjective retail judgement. The validity of Retail-CHAIR therefore depends directly on the quality of the L1–L4 ontology design and human annotation consistency — a limitation that will be explicitly acknowledged in the dissertation's evaluation chapter. **POPE** is adapted analogously: binary probing questions (e.g., "Is there a promotional shelf label in this image?", "Is the shelf fully stocked?") are derived directly from the L3/L4 ground-truth annotations and posed to the model as yes/no VQA queries, measuring object existence accuracy without requiring additional retail expertise. **LLM-as-Judge** (GPT-4o) provides supplementary hallucination verification across the full evaluation set as a cross-validation check on Retail-CHAIR scores; this metric is treated as indicative only given the GPT-4o circularity concern discussed in Section 5.2.

Stage 3 — Caption Quality (RQ3): CIDEr: evaluated against the 150–200-image human-reference subset. SPICE: scene-graph-based semantic evaluation, particularly suited to retail scenes requiring accurate object, attribute, and relation assessment [3].

6.2 Human Evaluation (Small-Scale)

A Retail Actionability Score (RAS) evaluation will be conducted with a panel of **5–10 fellow students**, assessing a random sample of **60–90 captions** (20–30 per ablation variant) on a 0–2 scale: 0 = Useless, 1 = Informative but non-actionable, 2 = Actionable (triggers a concrete operational task). Given the small evaluator panel, statistical comparisons will be limited to mean RAS with standard deviation, and significance testing (Wilcoxon signed-rank) will be applied where sample size permits. The limitations of small-panel evaluation will be explicitly discussed in the dissertation.

6.3 Ablation Study (Simplified)

Three experimental variants will be evaluated to isolate the contribution of the core pipeline components. A full multi-factor ablation is out of scope given time constraints; instead, the three variants are chosen to directly address the PRQ and RQ2.

Variant	Description	Purpose
Baseline A	LLaVA-1.5-7B with no tag constraints (pure end-to-end)	Establishes hallucination rate without any structural constraint
Baseline B	Flat tag injection only — Hard Tags listed in prompt, no routing, no absence encoding	Isolates the contribution of Routing + Absence Encoding
Proposed System	Full 3-stage pipeline: Hard/Soft tags, Routing, Absence Encoding, Self-Correction	Tests the full PRQ hypothesis

7. Project Timeline and Milestones

The project runs from late March 2026 to early August 2026 (approximately 18 weeks). A reduced-intensity period is incorporated for the May examination period. The scope has been calibrated to be achievable within this window while meeting MSc dissertation standards.

Phase	Period	Key Deliverables	Intensity
Phase 1: Foundation & Data	Late March – Late April	Environment setup (GPU, vLLM stack, CLIP, Grounding DINO, PaddleOCR); file-path metadata parser; image quality filtering; LLM-assisted annotation of 300–400 images (human-first tag review + GPT-4o suggestions); train/validation/test split	Full

	(Weeks 1–5)	definition; inter-rater reliability pilot (50 images); initial data statistics report	
Phase 2: Stage 1 Implementation	Late April – Early May (Weeks 5–7)	CLIP zero-shot L1 classification integration and evaluation; Grounding DINO L2 open-vocabulary fixture detection; CLIP zero-shot L3 product matching against GS1 GPC vocabulary (+ linear probe fallback if needed); PaddleOCR + CLIP L4 attribute classification; Automated Dynamic Dictionary pipeline; Stage 1 tag precision/recall baseline computed on validation subset	Full
Phase 3: Literature Review Writing (Exam Buffer)	May (Weeks 8–11)	Dissertation Chapters 1–3 drafted (Introduction, Related Work, Data); no heavy model training; code documentation; exam preparation	Reduced
Phase 4: Stage 2 & Stage 3 Implementation	Early June – Late June (Weeks 12–15)	K-Means/DBSCAN clustering on L1+L4 co-occurrence vectors; LLM persona assignment; Routing Matrix generation; Structured prompt assembly (MOP); Explicit Absence Encoding; Self-Correction loop; end-to-end pipeline integration test; **Chapters 4–5 (Methodology) written in parallel with implementation**	Full
Phase 5: Experiments & Evaluation	July (Weeks 16–18)	All 3 ablation variants run on held-out test set; Retail-CHAIR, POPE, CIDEr, SPICE scores computed; student panel RAS evaluation (60–90 captions); LLM-as-Judge supplementary scoring; Silhouette/DB Index analysis; qualitative error analysis and caption case studies; **Chapter 6 (Experiments & Results) written in parallel with evaluation runs**	Full
Phase 6: Dissertation Completion	Late July – Early August (Weeks 18–19)	Chapters 7–8 written (Discussion, Conclusion); Chapters 1–6 revised incorporating supervisor feedback cycles (minimum two feedback rounds); bibliography; code packaging and documentation; final submission	Full

8. Risk Assessment and Mitigation

Risk	Likelihood	Impact	Mitigation
Student tag annotation contains systematic	Mediu	Med	GPT-4o used as suggestion generator only (not final authority); researcher reviews every image directly

errors due to unfamiliarity with the L1–L4 ontology or annotator fatigue during visual review	m	ium	against the ontology schema; 50-image inter-rater reliability check with second student annotator; report Cohen's kappa; annotation guidelines documented and versioned in the dissertation appendix
Personal GPU insufficient for LLaVA-1.5-7B inference across 10,000 images	Medium	Medium	Use 4-bit quantisation (GPTQ/AWQ) for development; vLLM for batched throughput; request KCL HPC access for final evaluation runs
CLIP zero-shot precision on fine-grained L3 GS1 product categories is insufficient (e.g., failing to distinguish product sub-types within a GPC Brick)	High	Medium	Evaluate CLIP zero-shot precision on a 50-image pilot early in Phase 1; if precision falls below 60%, train a lightweight linear probe on frozen CLIP (ViT-L/14) embeddings of the 300–400 annotated images — data-efficient, requires no bounding-box annotations, feasible on a single GPU
Unsupervised clustering produces ambiguous or non-interpretable scenario groups	Medium	High	Evaluate K range 3–8; test both K-Means and DBSCAN; if clustering is unstable, document as a negative result and fall back to 3 manually seeded scenario types (Inventory, Safety/Compliance, Merchandising) — a finding in itself
May exam period extends and encroaches on Phase 4 start	Low	Medium	Phase 3 intentionally uses exam period for writing-only tasks (Chapters 1–3); Phase 4 implementation can begin immediately post-exam with no development progress lost
GPT-4o API costs for Dynamic Dictionary orchestration and LLM-as-Judge evaluation exceed budget	Low	Low	Use GPT-4o-mini for all orchestrator tasks; reserve full GPT-4o calls for final evaluation scoring only; batch API calls to minimise token usage; estimated total cost < £20–30
Small student evaluation panel (5–10) produces high variance in RAS scores	High	Low	Supplement RAS with LLM-as-Judge actionability scoring as a parallel automated metric; discuss small-panel limitations explicitly in the dissertation; prioritise quantitative Retail-CHAIR and POPE as the primary hallucination claims

9. Expected Contributions

This project makes the following anticipated contributions.

- **Methodological**: A novel three-stage data-driven tag-constrained captioning pipeline, introducing Explicit Absence Encoding and unsupervised tag-to-task routing as

structural hallucination mitigation mechanisms not present in prior image captioning literature [1], [2], [4], [5].
- **Domain**: A four-tier retail visual ontology combined with a file-path-metadata-driven Dynamic Dictionary, providing a practical and reusable tagging framework for retail vision research that requires no external metadata collection infrastructure.
- **Empirical**: A comparative ablation study on a real-world grocery retail image dataset demonstrating the individual and combined contributions of each pipeline component to hallucination reduction and caption actionability.
- **Practical**: A deployable, compute-efficient captioning pipeline (7B VLM, lightweight detectors) with GDPR-compliant anonymisation, suitable for deployment in UK grocery retail contexts.

10. Conclusion

This revised preliminary report incorporates five key practical adjustments relative to the initial proposal: (i) ground truth annotation is performed via an LLM-assisted semi-automated workflow with student review, removing the dependency on domain expertise; (ii) the Automated Dynamic Dictionary is redesigned to leverage the rich metadata embedded in the dataset's file-path structure (campaign type, brand, category, location), removing any reliance on external data sources; (iii) the project timeline has been restructured into six phases across approximately 18 weeks, with a dedicated reduced-intensity phase during May examinations and scope calibrated to MSc dissertation standards; (iv) the L1–L2 ontology vocabulary is derived via a two-pass data-driven induction process — GPT-4o Vision analysis of a stratified 100-image corpus sample followed by alignment with retail visual merchandising literature (L2) and GS1 GPC Brick nomenclature (L3) — removing any dependency on prior retail domain knowledge in vocabulary design, with the L4 operational state attributes derived directly from the project's research questions; and (v) the generative VLM is downsized to a 7B-parameter model to remain within personal GPU constraints, with KCL HPC reserved for final evaluation. The core research contribution — a data-driven pipeline that structurally constrains VLM generation through hierarchical tagging, unsupervised scenario routing, and explicit absence encoding — remains intact and is positioned as a novel and empirically grounded approach to the hallucination problem in retail image captioning.

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